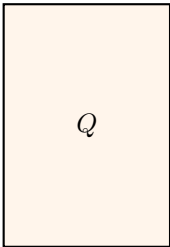
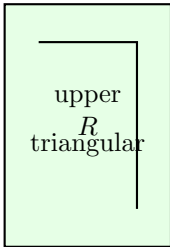
 $A$  $=$  $Q$ 

upper  
 $R$   
triangular

$Q^*Q = \mathcal{I}$  coordinates after orthogonalization